

Contents

1	Introduction	1
2	Exploratory Analysis of Ranking Data	7
2.1	Descriptive Statistics	7
2.2	Visualizing Ranking Data	9
2.2.1	Permutation Polytope	9
2.2.2	Multidimensional Scaling and Unfolding	11
2.2.3	Multidimensional Preference Analysis	15
2.2.4	Comparison of the Methods	19
2.3	Tests for Randomness on Ranking of a Set of Objects	19
	Chapter Notes	21
3	Correlation Analysis of Paired Ranking Data	23
3.1	Notion of Distance Between Two Rankings	23
3.2	Correlation Between Two Rankings	26
3.3	Incomplete Rankings and the Notion of Compatibility	29
3.4	Correlation for Incomplete Rankings	32
3.4.1	Asymptotic Normality of the Spearman and Kendall Test Statistics	36
3.4.2	Asymptotic Efficiency	41
3.5	Tied Rankings and the Notion of Compatibility	43
3.6	Angular Correlations	45
3.6.1	Spearman Distance	47
3.6.2	Kendall Distance	48
3.6.3	Asymptotic Distributions	49
3.7	Angle-Linear Correlation	50
	Chapter Notes	52
4	Testing for Randomness, Agreement, and Interaction	55
4.1	Tests for Randomness	55
4.2	Tests for Agreement Among Groups	64

4.3	Test for Interaction in a Two-Way Layout	70
4.3.1	Proposed Row–Column Test Statistic	72
4.3.2	Asymptotic Distribution of the Test Statistic Under the Null Hypothesis	73
	Chapter Notes	78
5	Block Designs	81
5.1	Incomplete Block Designs	81
5.2	Asymptotic Distribution	85
5.3	Spearman Case	85
5.3.1	Applications	88
5.4	Kendall Case	91
5.5	Hamming Case	94
5.5.1	Applications	97
5.6	Score Statistics	98
5.6.1	Special Score Functions	103
	Chapter Notes	104
6	General Theory of Hypothesis Testing	105
6.1	The Construction	105
6.2	The Multi-Sample Location Problem with Ordered Alternatives	106
6.2.1	Asymptotic Distribution Under the Alternative	110
6.3	Tests Under Umbrella Alternatives	113
6.3.1	The Construction of the Test Statistics	114
6.3.2	The Test Statistic Corresponding to Spearman Distance	116
6.3.3	The Test Statistic Corresponding to Kendall Distance ...	118
6.3.4	The Asymptotic Distribution of the Test Statistics Under the Null Hypothesis	120
6.3.5	The Test Statistics When the Location of the Peak is Unknown	122
6.3.6	Simulation Study	123
	Chapter Notes	125
7	Testing for Ordered Alternatives	127
7.1	Test Statistics When the Data Is Complete in Each Block	128
7.2	The Incomplete Case	132
7.2.1	Asymptotic Efficiency	137
7.3	Tests for Trend in Proportions	137
7.3.1	The Construction of the Test Statistics	140
7.3.2	The Test Statistic Corresponding to Spearman and Kendall Similarity	141

7.3.3	The Test Statistic Corresponding to Hamming Similarity	142
7.3.4	The Asymptotic Distribution of the Test Statistics Under the Null Hypothesis	143
	Chapter Notes	146
8	Probability Models for Ranking Data	149
8.1	Order Statistics Models	150
8.1.1	Luce Model.....	151
8.1.2	Rank-Ordered Logit Models.....	153
8.1.3	Some Non-IIA Order Statistics Models.....	155
8.2	Paired Comparison Models	159
8.3	Distance-Based Models	160
8.3.1	ϕ -Component Models	162
8.3.2	Cyclic Structure Models	164
8.4	Multistage Models	165
8.5	Properties of Ranking Models	167
8.5.1	Properties of Order Statistics Models	167
8.5.2	Properties of Paired Comparison Models	168
8.5.3	Properties of Distance-Based and Multistage Models....	168
	Chapter Notes	169
9	Probit Models for Ranking Data	171
9.1	Multivariate Normal Order Statistics Models	172
9.1.1	The MVNOS Model with Covariates	172
9.1.2	Parameter Identifiability of the MVNOS Model	173
9.1.3	Bayesian Analysis of the MVNOS Model	175
9.1.4	Adequacy of the Model Fit	178
9.1.5	Analysis of the APA Election Data	179
9.2	Factor Analysis	183
9.2.1	The Factor Model	183
9.2.2	Monte Carlo Expectation-Maximization Algorithm	185
9.2.3	Simulation	187
9.2.4	Factor Score Estimation	189
9.2.5	Application to the Job Selection Ranking Data.....	191
	Chapter Notes	197
10	Decision Tree Models for Ranking Data	199
10.1	Impurity Function Approach	201
10.1.1	Building Decision Tree for Ranking Data	201
10.1.2	Leaf Assignment	203
10.1.3	Performance Assessment of Decision Tree for Ranking Data	204
10.1.4	Analysis of 1993 European Value Priority Data.....	205

10.2	Statistical Test Approach Based on Intergroup Concordance	211
10.2.1	Building Decision Tree Using Test for Intergroup Concordance	211
10.2.2	Analysis of US General Social Survey Data on Job Value Preference	212
11	Extension of Distance-Based Models for Ranking Data	223
11.1	Weighted Distance-Based Models	223
11.1.1	Properties of Weighted Distance-Based Models.....	224
11.2	Mixtures of Weighted Distance-Based Models	226
11.2.1	Analysis of Croon's Political Goals Data	228
11.3	Distance-Based Tree Models	233
11.3.1	Building Distance-Based Tree Models.....	233
11.3.2	Analysis of 1999 European Value Priority Data.....	234
	Chapter Notes	237
	Appendix A: Ranking Data Sets	239
A.1	Goldberg Data	239
A.2	Leisure Time Data	240
A.3	Data on the Big Four EPL Teams	240
A.4	MovieLens Data	241
A.5	Language and Arithmetic Scores Data	241
A.6	Public Opinion Survey Data.....	241
A.7	Wind Direction Data.....	242
A.8	Lymph Heart Pressure Data	242
A.9	Mortality Data for South Africa 2000–2008	243
A.10	<i>E. coli</i> Data for Six Beaches in Hong Kong	243
A.11	Umbrella Alternative Data	244
A.12	APA Election Data.....	244
A.13	Job Selection Data	244
A.14	1993 European Value Priority Data	245
A.15	US General Social Survey Data	246
A.16	Song Data	247
A.17	Croon's Political Goals Data	247
A.18	1999 European Value Priority Data	248
	Appendix B: Limit Theorems	251
B.1	Hoeffding's Combinatorial Central Limit Theorem	251
B.2	Multivariate Central Limit Theorem	252
B.3	Quadratic Forms	252
B.4	Asymptotic Efficiency	253
B.4.1	One-Sided Tests	253

Appendix C: Review on Decision Trees	255
C.1 Introduction	255
C.2 CART Algorithm	255
C.2.1 Growing Stage of Decision Tree	255
C.2.2 Pruning Stage of Decision Tree	257
C.2.3 Class Assignment of Leaf Nodes of Decision Tree	259
C.3 Other Decision Tree Models	259
Bibliography	261
Index	271